

PERPLEXITY IN MARKOV N-GRAM MODELS

$$\begin{aligned} PP(w_1^n) &= P(w_1^n)^{-\frac{1}{n}} \\ &= \sqrt[n]{\frac{1}{P(w_1^n)}} \\ &= \sqrt[n]{\prod_{i=1}^n \frac{1}{P(w_i | w_1^{i-1})}} \end{aligned}$$

Causes overflow

$$= \sqrt[n]{\frac{1}{\prod_{i=1}^n P(w_i | w_1^{i-1})}}$$

Causes underflow

Hmm what to do now o_O!?

Let's now examine how N -gram language models are represented. We represent and compute language model probabilities in log format, in order to avoid underflow and also to speed up computation. Since probabilities are (by definition) less than 1, the more probabilities we multiply together the smaller the product becomes. Multiplying enough N -grams together would result in numerical underflow. By using log probabilities instead of raw probabilities, the numbers are not as small. Since adding in log space is equivalent to multiplying in linear space, we combine log probabilities by adding them. Besides avoiding underflow, addition is faster to compute than multiplication.

<http://www.cs.colorado.edu/%7Emartin/SLP/Updates/4.pdf> Seite 26

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^^ So let's switch to logarithmic space for a better distribution!

1) Multiplication equals summation in logarithmic space:

$$\prod_{i=1}^n x_i = \exp\left(\sum_{i=1}^n \ln x_i\right)$$

2) n-th root of x is its exponentiation to 1/n:

$$\sqrt[n]{x} = x^{\frac{1}{n}}$$

3) Exponentiation equals inverse multiplication in logarithmic space:

$$a^b = \exp(b \ln a)$$

Now apply these three equations to the raw perplexity formulae ...

$$PP(w_1^n) = P(w_1^n)^{-\frac{1}{n}}$$
$$PP(w_1^n) = \sqrt[n]{\prod_{i=1}^n \frac{1}{P(w_i | w_1^{i-1})}}$$

... and here they are: two formulae that avoid under-, over-flows:

$$PP(w_1^n) = \exp\left(\frac{1}{n} \sum_{i=1}^n \ln \frac{1}{P(w_i | w_1^{i-1})}\right)$$
$$PP(w_1^n) = \exp\left(-\frac{1}{n} \sum_{i=1}^n \ln P(w_i | w_1^{i-1})\right)$$

This is in libcorisis since SVN revision 389.

$$\sqrt[n]{\prod_{i=1}^n \frac{1}{P(w_i|w_1^{i-1})}} = \left(\prod_{i=1}^n P(w_i|w_1^{i-1})\right)^{\frac{1}{n}}$$

$$= \exp\left(-\frac{1}{n}\sum_{i=1}^n \ln P(w_i|w_1^{i-1})\right)$$

$$E_i = \frac{N_i \sum_i O_i}{\sum_i N_i}$$

$$-2\ln\lambda = 2\sum_i o_i \ln\Big(\frac{O_i}{E_i}\Big) \quad \square \quad \square$$